# Data Anonymisation with the Density Matrix Classifier

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**Abstract**

We propose a new data anonymisation method based on the concept of a quantum feature map. The main advantage of the proposed solution is that a high degree of security is combined with the ability to perform classification tasks directly on the anonymised (encrypted) data resulting in the same or even higher accuracy compared to that obtained when working with the original plain text data. This enables important usecases in medicine and finance where anonymised datasets from different organisations can be combined to facilitate improved machine learning outcomes utilising the combined dataset. Examples include combining medical diagnostic imaging results across hospitals, or combining fraud detection datasets across financial institutions. We use the Wisconsin Breast Cancer dataset to obtain results on Rigetti's quantum simulator and Ankaa-3 quantum processor. We compare the results with classical benchmarks and with those obtained from an alternative anonymisation approach using a Restricted Boltzmann Machine to generate synthetic datasets. Finally, we introduce concepts from the theory of quantum magic to optimise the circuit ansatz and hyperparameters used within the quantum feature map.

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# 1 Introduction

Data anonymisation and, more broadly, data privacy are fundamental problems we are facing in the digital age. Often data will not be shared or disclosed without guarantees that sensitive information cannot be extracted. Be it client data of financial institutions or patient data in a medical context, such highly sensitive data usually cannot be shared externally. In many cases, even internal usage of sensitive data is done on a "need to know" basis. There is a strong demand for reliable and secure data anonymisation solutions.

Traditional anonymisation methods include notions such as data de-identification [1] (removing attributes of individuals or substituting artificial values, or by releasing only fractions of an anonymised dataset) and perturbation methods [2] (common forms of perturbation are adding noise, data transformation and rotation). A number of models have been proposed that can be used to obtain a definable amount of privacy. An example is the well-known $k$-anonymity model [3, 4] that ensures that no private information can be related to fewer than $k$ individuals. A lack of diversity in such $k$-sets can still impede privacy though, and it may become possible to link private information to individuals. Improving on $k$-anonymity, the required critical diversity can be ensured using the $l$-diversity model [5]. However, the method can only ensure diversity for one sensitive attribute. Further research led to the development of $t$-closeness [6] and differential privacy [7]. However, the traditional anonymisation techniques are not effective on high-dimensional data. Indeed, the advances of machine learning applications have also improved our ability to reverse engineer de-identified datasets in order to trace information back to specific individuals [8].

A novel approach to data anonymisation is offered by generative models based on deep neural networks. This utilises data generation, that is, the generation of new data instead of altering the original dataset. Data anonymisation solutions based on the Generative Adversarial Network (GAN) [9, 10] and the Restricted Boltzmann Machine (RBM) [11] have been applied to medical datasets. Generative model-based solutions preserve the data generating distribution by training either on the original data or on their accurate representations and can in turn be used to generate new data with the same characteristics as the original data. This approach is by construction safer then data perturbation, as newly generated data points cannot be identified with original data points, where the quality-privacy trade-off can be balanced better than in traditional methods.

This paper explores an alternative approach. It addresses the problem of data anonymisation with the help of Quantum Machine Learning (QML) techniques that have attracted much interest in recent years [12, 13, 14]. As quantum computing technology matures, it becomes feasible to experiment with the models and algorithms supported by this new computational paradigm. The paper investigates two specific QML models that can be used for data anonymisation purposes: the Quantum Feature Map (QFM) and the Density Matrix Classifier (DMC). The QFM performs the data anonymisation (or data encryption) task by mapping every sample in the dataset into a corresponding pure quantum state. The DMC then utilises the fact that all samples corresponding to a particular class can be combined into a mixed quantum state (a statistical ensemble of pure quantum states) that can be described with the help of a density matrix. The Frobenius distance between the density matrices can then be used as a measure of similarity of the subsets (which each consist of samples belonging to a specific class). Similarly, every new sample can be classified by calculating the Frobenius distance between its own density matrix and the density matrices of all class subsets in the dataset. The new sample is assigned a class label corresponding to the class with the smallest Frobenius distance. Therefore, we have a powerful native classifier that can operate on the fully anonymised data without the need for decryption. In a practical context, this means that dataset owners can anonymise their data, amalgamate it across owners, and use the combined dataset for developing classifiers. It can result in improved classification accuracy compared to classifiers trained on only the dataset of a single owner.

**Paper contributions:** The contributions made in this paper are:

- Presentation of a new algorithm utilising QML techniques to provide data anonymisation (encryption) and the subsequent classification of anonymised (encrypted) data.
- Application of the algorithm to anonymise and classify the Wisconsin Breast Cancer dataset.
- Presentation of the results obtained on Rigetti's quantum simulator and the Ankaa-3 quantum processor.
- Comparison of results with those obtained from a Restricted Boltzmann Machine synthetic data generator model.
- Demonstration of how the theory of quantum magic can be used to improve the discrimination capabilities of a QFM and optimise the circuit ansatz and hyperparameters.

The paper is organised as follows. In Section 2 we introduce the concepts of QFM and DMC. In Section 3 we present the proposed data anonymisation protocol. In Section 4 we illustrate performance of the DMC-based data anonymisation protocol and related classification task on the Wisconsin Breast Cancer dataset [15]. In Section 5 we provide comparison with an alternative data anonymisation method based on a synthetic data generator. Section 6 concludes with the discussion on the achievable level of data privacy and outlines the programme of future work. The appendices provide additional information on the QFM and data encoding scheme used. The classical benchmarks are described. Algorithms for estimating the Frobenius distance between datasets and implementing RBMs are presented. Key concepts of magic in parameterised quantum circuits are summarised.

# 2 Quantum Feature Map and Density Matrix Classifier

For those unfamiliar with the field of quantum computing, an introduction to the terminology and concepts is provided in [16].

## 2.1 Quantum Feature Map

Let $\mathcal{D}$ be a dataset consisting of samples belonging to $N$ non-overlapping subsets (classes) $\mathcal{C}_1, \ldots, \mathcal{C}_N$:

$$\mathcal{C}_1 \cup \ldots \cup \mathcal{C}_N = \mathcal{D}, \quad \mathcal{C}_i \cap \mathcal{C}_j = \emptyset \quad \forall\, i \neq j\ . \tag{1}$$

Each sample in the dataset, $\mathbf{x} \in \mathcal{D}$, can be represented as a vector of features, $\mathbf{x} := (x_1, \ldots, x_m)$, where $m$ is the number of features, or the *dimensionality* of the dataset. We define the feature map $\varphi$ as a function that maps samples $\mathbf{x}$ into operators $\varphi(\mathbf{x})$ of a Hilbert space $\mathcal{H}$. The feature map $\varphi$ is realised on a quantum computer via a unitary transformation $U(\mathbf{x})$ applied to the initial state $|\mathbf{0}\rangle \equiv |0\rangle^{\otimes n}$ [16]:

$$\varphi(\mathbf{x}) := U(\mathbf{x})\,|\mathbf{0}\rangle\,\langle\mathbf{0}|\,U(\mathbf{x})^\dagger \equiv |\mathbf{x}\rangle\,\langle\mathbf{x}|\,. \tag{2}$$

Note that if the transformation $\varphi$ is specified on $n$ qubits, then the dimension of the Hilbert space $\mathcal{H}$ of linear operators is $4^n = 2^n \times 2^n$.

The quantum state $|\mathbf{x}\rangle := U(\mathbf{x})\,|\mathbf{0}\rangle$ is a *pure* quantum state encoding sample $\mathbf{x}$, and the feature map $\varphi(\mathbf{x})$ is the corresponding density matrix. The density matrix formalism allows us to describe not only pure but also *mixed* quantum states, i.e., statistical ensembles of pure quantum states. Thus, the density matrix approach can be used to encode not just a single sample but the whole subset of samples belonging to a particular class.

For each class $\mathcal{C}_i \subset \mathcal{D}$, $i = 1, \ldots, N$, we construct a density matrix $\rho_i$ defined as an equally weighted statistical ensemble of all samples in $\mathcal{C}_i$:

$$\rho_i = \frac{1}{[\mathcal{C}_i]} \sum_{k=1}^{[\mathcal{C}_i]} |\mathbf{x}^k\rangle\,\langle\mathbf{x}^k|\,, \tag{3}$$

where $[\mathcal{C}_i]$ denotes the cardinality of $\mathcal{C}_i$ and the sum goes over all samples $\mathbf{x}^k, k = 1, \ldots, [\mathcal{C}_i]$ in class $\mathcal{C}_i$.

We can estimate the *closeness* of any two classes, $\mathcal{C}_i$ and $\mathcal{C}_j$, as the distance between the corresponding density matrices $\rho_i$ and $\rho_j$. The metric of choice is the Frobenius distance:

$$d_F(\rho_i, \rho_j) = \frac{1}{2}\sqrt{\mathrm{Tr}[(\rho_i - \rho_j)^2]}\ . \tag{4}$$

The primary motivation for choosing the Frobenius distance over other alternatives is that it can be efficiently estimated on a quantum computer [16] by calculating the expectation values of the first order Pauli operators ($\langle X\rangle$, $\langle Y\rangle$, and $\langle Z\rangle$) on all qubits for all samples in the dataset (see Appendix D for details).

## 2.2 Density Matrix Classifier

The DMC can be formulated as the following algorithm [17]:

**Algorithm 1** Density Matrix Classifier

**Result:** Classification of a data sample.

1: Start with the dataset consisting of $N$ non-overlapping subsets (classes) $\mathcal{C}_1, \ldots, \mathcal{C}_N$ as described in (1). Each sample $\mathbf{x}$ in the dataset is an $m$-dimensional vector of real-valued features. Discrete or categorical variables can also be represented.

2: Specify an $n$-qubit quantum circuit that implements the unitary transformation $U(\mathbf{x})$ applied to the initial all-zero state $|0\rangle^{\otimes n}$. For example, the quantum circuit may consist of several layers of adjustable 1-qubit gates encoding components of $\mathbf{x}$ into qubit rotation angles, and layers of fixed 2-qubit gates creating entanglement.

3: For each class $\mathcal{C}_i, i = 1, \ldots, N$, and for each sample in class $\mathcal{C}_i$, $|\mathbf{x}^k\rangle, k = 1, \ldots, [\mathcal{C}_i]$, construct the feature $\varphi(\mathbf{x}^k)$ given by expression (2) using the unitary transformation $U(\mathbf{x}^k)$.

4: For each class $\mathcal{C}_i$, $i = 1, \ldots, N$, construct a density matrix $\rho_i$ defined as an equally weighted statistical ensemble of all samples in $\mathcal{C}_i$ as per expression (3).

5: For each new sample to be classified, $\mathbf{x}$, construct its own feature map $\varphi(\mathbf{x})$ and density matrix $\rho$ using the unitary transformation $U(\mathbf{x})$. For each class $\mathcal{C}_i, i = 1, \ldots, N$, calculate the Frobenius distance $d_F(\rho_i, \rho)$ given by expression (4). The new sample is assigned a class label corresponding to the class with the smallest Frobenius distance.

It appears that we can get a sufficiently accurate estimate of the Frobenius distance without constructing the density matrices explicitly. The Frobenius distance estimator can be constructed from the expectation values of all first order Pauli operators calculated for all samples in the dataset on all qubits in the quantum circuit that implements the QFM as described in [17].

The anonymisation (encryption) of a data sample therefore consists of mapping it into a quantum state with the help of a QFM and then calculating $\langle X \rangle$, $\langle Y \rangle$, and $\langle Z \rangle$ by running the corresponding quantum circuit multiple times and performing the measurements on all qubits in the $x$, $y$, and $z$ bases. Thus, the original $m$-feature sample $\mathbf{x} := (x_1, \ldots, x_m)$ will be mapped into a $3n$-dimensional vector of expectation values of all first order Pauli operators $(\langle X_1 \rangle, \ldots, \langle X_n \rangle, \langle Y_1 \rangle, \ldots, \langle Y_n \rangle, \langle Z_1 \rangle, \ldots, \langle Z_n \rangle)$, where $n$ is the number of qubits in the quantum circuit implementing the QFM.

# 3 Data Anonymisation Protocol

**Hypothetical example:** Healthcare providers A, B, and C would like to combine their medical records to create a larger dataset in order to build a better ML classifier for diagnostic purposes. However, they are facing serious data privacy constraints.

**Possible solution:** A, B, and C can anonymise their datasets using one of the standard data anonymisation techniques (differential privacy, etc.) and perform classification on the combined anonymised data.

**Suggestion:** A, B, and C can do better than that. The recently proposed applications utilising parameterised quantum circuits (PQCs) executable on NISQ computers [16, 17, 18] allow us to achieve full data anonymisation with the following simultaneous operations:

i) quantum feature map;

ii) quantum encryption.

The quantum feature map enhances the classifier performance while quantum encryption provides data privacy.

**Proposed protocol:**

1. A, B, and C choose a suitable PQC that can encode each sample (vector of features) into the corresponding pure quantum state, thus realising the QFM. For example, the PQC may consist of layers of 1-qubit gates encoding the features and 2-qubit gates creating entanglement – a standard PQC ansatz.
2. However, quantum states cannot be easily shared or stored in memory. Therefore, A, B, and C must perform the following operation. For each sample, the PQC is run multiple times with the measurements on each qubit performed in the $x$, $y$, and $z$ bases. The resulting bitstrings are used to calculate the averages – the expectation values of the corresponding Pauli operators $X$, $Y$, and $Z$ on every qubit. Thus, each sample is mapped into a vector of expectation values of all possible first order Pauli operators.
3. The categorical class labels are mapped directly into the corresponding integer values.
4. A, B, and C send encrypted datasets to the data aggregator D. The aggregator combines them into a single dataset on which a classifier can be trained. D has no ability to decrypt the encrypted data, even in the case it learns the PQC ansatz.
5. We have reached the point where we can directly use the Density Matrix Classifier as described in [17].
6. When any of the healthcare providers would like to classify a new sample, they need to anonymise/encrypt it with the help of the same PQC and send the encrypted sample to D for classification. D will process the sample and return the assigned class label (a single integer number).

**Key observations:**

- The dataset is fully anonymised and encrypted – it is believed to be impossible to perform the inverse operation and recover the vector of features from the vector of expectation values of first order Paulis even when the PQC ansatz is known (for a non-trivial PQC ansatz).

- Classification is performed on encrypted data.
- The quantum feature map itself can improve classification results due to the large expressive power of parameterised quantum circuits.

# 4 Numerical Experiments

We illustrate the implementation of the proposed data anonymisation protocol on the popular medical Wisconsin Breast Cancer dataset. The central idea is to split it into three equal subsets (representing the hypothetical healthcare providers A, B, and C), perform data anonymisation with the help of a QFM, combine anonymised (encrypted) subsets into a single dataset, and perform classification of new unseen samples (out-of-sample classification) directly on the encrypted data.

## 4.1 Wisconsin Breast Cancer Dataset

The dataset [15] consists of 683 samples (after all samples with missing features have been removed). There are nine integer features and the classification problem is binary: 444 samples are labeled as "benign" and 239 samples are labeled as "malignant". Feature importance analysis conducted with the help of a random forest classifier suggests the ranking of features in terms of their relative predictive power as shown in Table 1.

| Feature | Importance |
|---|---|
| Uniformity of Cell Size (F2) | 26.3% |
| Uniformity of Cell Shape (F3) | 21.7% |
| Bare Nuclei (F6) | 18.3% |
| Bland Chromatin (F7) | 10.3% |
| Single Epithelial Cell Size (F5) | 8.2% |
| Normal Nucleoli (F8) | 7.1% |
| Clump Thickness (F1) | 4.8% |
| Marginal Adhesion (F4) | 2.5% |
| Mitoses (F9) | 0.7% |

Table 1: Feature importance for the Wisconsin Breast Cancer dataset, as determined by a random forest classifier.

## 4.2 Data Anonymisation with Quantum Feature Map

The quantum circuit implementing the QFM and its embedding onto the QPU graph are shown Figures 5 and 6 (Appendix A). The quantum circuit consists of nine qubits entangled with the help of three layers of 2-qubit $iSWAP$ gates. Three layers of 1-qubit gates perform repeat angle encoding of the original nine features as explained in Appendix B. The QPU graph is a $3 \times 3$ square grid of computational qubits connected by tunable couplers. This graph can be represented by a section of

Rigetti's 84-qubit Ankaa-3 system, or a full 9-qubit Novera system [19] – a 9-qubit version of the Ankaa architecture.

Table 2 shows six randomly selected samples from classes "0" (benign) and "1" (malignant). All nine features are integer numbers between 1 and 10.

| Feature | Sample 1 | Sample 2 | Sample 3 | Sample 4 | Sample 5 | Sample 6 |
|---|---|---|---|---|---|---|
| F1 | 3 | 3 | 2 | 5 | 4 | 4 |
| F2 | 1 | 1 | 1 | 10 | 8 | 8 |
| F3 | 1 | 1 | 1 | 10 | 6 | 8 |
| F4 | 1 | 1 | 1 | 3 | 4 | 5 |
| F5 | 2 | 3 | 2 | 7 | 3 | 4 |
| F6 | 1 | 2 | 1 | 3 | 4 | 5 |
| F7 | 2 | 1 | 1 | 8 | 10 | 10 |
| F8 | 1 | 1 | 1 | 10 | 6 | 4 |
| F9 | 2 | 1 | 1 | 2 | 1 | 1 |
| Class | 0 | 0 | 0 | 1 | 1 | 1 |

Table 2: Six randomly selected samples from the Wisconsin Breast Cancer Dataset with one half belonging to class "0" (benign) and one half belonging to class "1" (malignant).

Each 9-feature sample is transformed into a 27-dimensional vector of expectation values of first order Pauli operators. Table 3 shows samples 1–6 in their anonymised (encrypted) form. There are several considerations that ensure that the original (plain text) form of the samples cannot be deduced from any number of encrypted samples.

First, knowledge of the expectation values of the first order Pauli operators is insufficient to learn exactly the quantum state. To reconstruct the density matrix describing a particular quantum state we need to know the expectation values of higher order Paulis [20].

Second, knowledge of the quantum state (or a collection of quantum states) is insufficient to reconstruct the quantum circuit used to create it [21, 22].

Third, the expectation values of the first order Pauli operators are known with finite precision due to the inherently probabilistic nature of the measurement process and the finite number of quantum circuit runs.

Fourth, even if the quantum circuit becomes known to the adversary, it implements a one-way function and cannot be used to invert the vector of expectation values of first order Paulis to reconstruct the original features with a high degree of confidence [23].

| Feature | Sample 1 | Sample 2 | Sample 3 | Sample 4 | Sample 5 | Sample 6 |
|---|---|---|---|---|---|---|
| $\langle X\rangle$, qubit 1 | 0.9461 | 0.9904 | 0.9975 | 0.2952 | -0.0034 | 0.0457 |
| $\langle X\rangle$, qubit 2 | -0.0026 | -0.0059 | 0.0055 | 0.0827 | 0.0259 | -0.0270 |
| $\langle X\rangle$, qubit 3 | 0.0136 | 0.0017 | -0.0061 | 0.1035 | 0.1017 | -0.0224 |
| $\langle X\rangle$, qubit 4 | -0.0067 | 0.2623 | 0.0020 | -0.0128 | -0.0114 | -0.0653 |
| $\langle X\rangle$, qubit 5 | 0.0014 | 0.0049 | -0.0063 | 0.0977 | 0.0362 | 0.1100 |
| $\langle X\rangle$, qubit 6 | -0.0023 | 0.0051 | 0.0154 | 0.0490 | -0.0270 | 0.0169 |
| $\langle X\rangle$, qubit 7 | -0.0194 | -0.0074 | 0.0055 | -0.0338 | -0.2068 | -0.1493 |
| $\langle X\rangle$, qubit 8 | 0.0045 | -0.0003 | -0.0070 | -0.0119 | 0.0615 | 0.0850 |
| $\langle X\rangle$, qubit 9 | 0.9177 | 0.9578 | 1.0000 | -0.3410 | -0.1370 | -0.1467 |
| $\langle Y\rangle$, qubit 1 | 0.0068 | 0.0363 | 0.0092 | -0.0080 | 0.0044 | -0.1100 |
| $\langle Y\rangle$, qubit 2 | 0.0188 | -0.0087 | 0.0035 | -0.0846 | 0.1357 | 0.0310 |
| $\langle Y\rangle$, qubit 3 | 0.0032 | 0.0025 | 0.0102 | -0.0567 | -0.0305 | 0.0071 |
| $\langle Y\rangle$, qubit 4 | 0.0108 | -0.0066 | -0.0050 | -0.0279 | 0.0028 | -0.0153 |
| $\langle Y\rangle$, qubit 5 | 0.0032 | 0.0039 | -0.0049 | -0.3047 | -0.2117 | -0.1102 |
| $\langle Y\rangle$, qubit 6 | 0.0027 | -0.0019 | -0.0117 | 0.0066 | -0.0001 | 0.0381 |
| $\langle Y\rangle$, qubit 7 | -0.0056 | 0.0002 | -0.0042 | 0.2039 | 0.0479 | 0.0355 |
| $\langle Y\rangle$, qubit 8 | -0.0083 | -0.0004 | -0.0017 | -0.0791 | 0.0339 | 0.0632 |
| $\langle Y\rangle$, qubit 9 | -0.0006 | -0.0112 | -0.0007 | 0.0060 | -0.0034 | -0.0615 |
| $\langle Z\rangle$, qubit 1 | 0.2850 | 0.0199 | -0.0004 | 0.6626 | 0.0916 | 0.0115 |
| $\langle Z\rangle$, qubit 2 | 0.5593 | 0.9036 | 0.5393 | -0.0015 | 0.2479 | 0.3846 |
| $\langle Z\rangle$, qubit 3 | 0.0064 | 0.2856 | -0.0040 | -0.3611 | -0.1872 | -0.3563 |
| $\langle Z\rangle$, qubit 4 | -0.0002 | -0.0134 | -0.0091 | -0.1491 | -0.5068 | -0.2049 |
| $\langle Z\rangle$, qubit 5 | 0.0081 | -0.0112 | 0.0063 | 0.0180 | 0.0735 | 0.2461 |
| $\langle Z\rangle$, qubit 6 | 0.2727 | 0.0027 | -0.0076 | 0.1359 | 0.0072 | -0.0143 |
| $\langle Z\rangle$, qubit 7 | 0.1443 | 0.1391 | 0.0638 | -0.2415 | 0.0139 | -0.1017 |
| $\langle Z\rangle$, qubit 8 | 0.0094 | 0.0105 | -0.0041 | 0.5074 | 0.4900 | 0.3605 |
| $\langle Z\rangle$, qubit 9 | 0.0022 | 0.0015 | -0.0014 | -0.3003 | 0.0951 | -0.3243 |
| Class | 0 | 0 | 0 | 1 | 1 | 1 |

Table 3: Transformed (anonymised) features. New features (expectation values of Pauli operators $X$, $Y$, and $Z$) were obtained by running the quantum circuit that implements the quantum feature map as described in Appendix A.

## 4.3 Classification Results

We can clearly solve the data anonymisation problem by performing the QFM, but do we risk losing valuable information during the data transformation process? Do the vectors of expectation values of first order Pauli operators contain sufficient information for sample classification purposes?

To answer these questions we performed the following numerical experiments. We randomly split the dataset into three roughly equal subsets with the same ratio of benign/malignant samples (stratified random split). We denote these subsets as datasets A, B, and C representing three hypothetical healthcare providers. Then we split each of these datasets into their corresponding train and test sets in an

80:20 proportion using the scikit-learn train_test_split function [24] with ten different values of the random_state parameter (0 to 9). For each split we train four standard classical classifiers on the train set and thereafter calculate their F1 classification scores on the test set. The results are shown as orange dots and bars in Figure 1. The average mean F1 score is just below 96% for the fully optimised classifiers (see Appendix C).

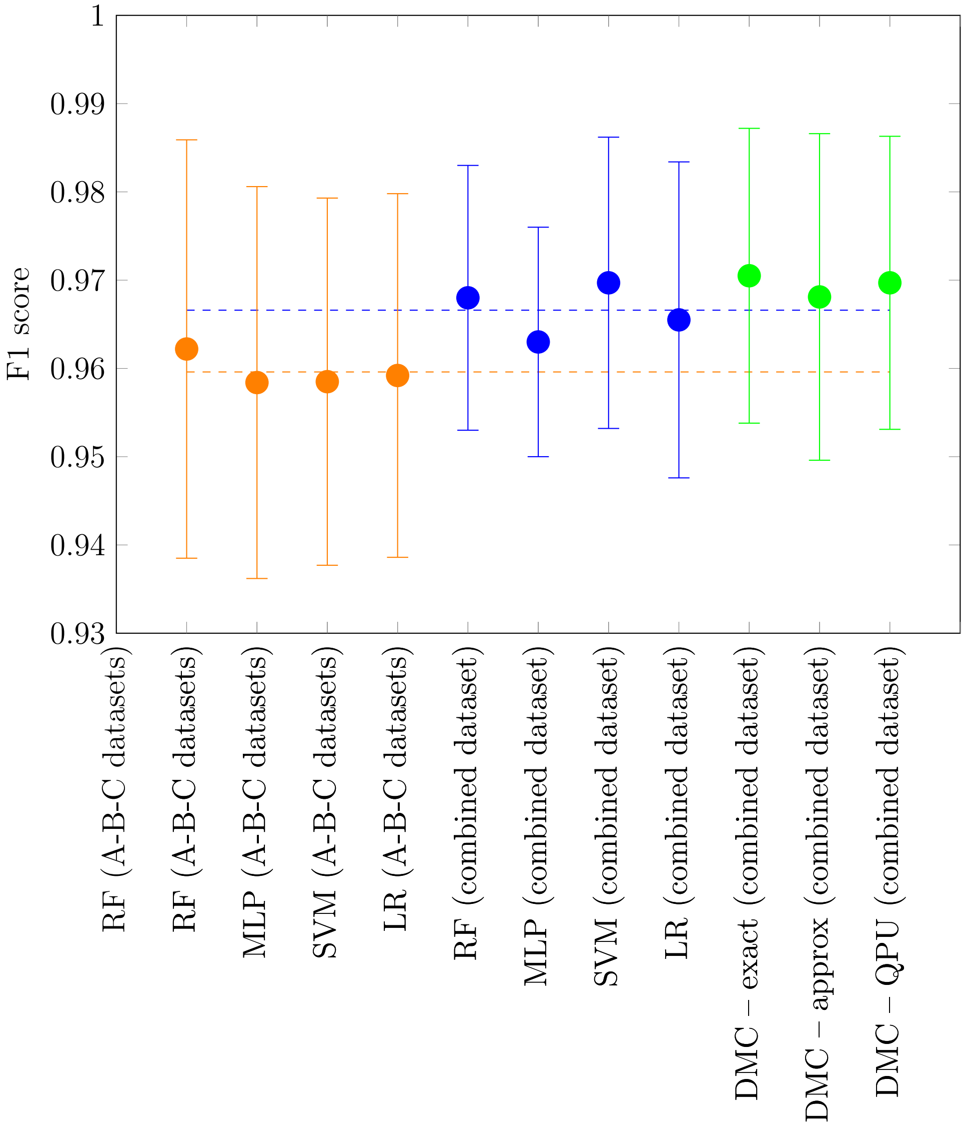


Figure 1: Out-of-sample F1 scores for classical (shown in orange and blue) and quantum (shown in green) classifiers using the Wisconsin Breast Cancer dataset. Dots indicate the mean values and error bars indicate ±1 standard deviations. All F1 scores were calculated with "macro" averaging across "benign" and "malignant" classes. The orange and blue dashed lines indicate the average of all the mean F1 score values across all classical classifiers for the full dataset (blue) and randomly split into three equal subsets (orange). The DMC results were obtained on Rigetti's quantum simulator (Rigetti QVM using pyQuil [25]) as well as the latest generation Rigetti quantum processor (Rigetti Ankaa-3 QPU [26]).

We repeated the process for the combined dataset with the classical out-of-sample F1 scores shown in Figure 1 as blue dots and bars. As expected, the results were improved (by 0.7% in terms of the average mean F1 score) due to the larger training set. This illustrates the benefit of working with the larger datasets and provides strong motivation for pooling the data.

Finally, we performed data anonymisation using the QFM described in Appendix A and the angle encoding scheme specified in Appendix B for the subsets A, B, and C. The anonymised samples were then combined into a single dataset. An 80:20 split into training and test sets was implemented. Classification using the DMC directly on the anonymised data (without decryption) was performed. The resulting F1 score statistics are shown in Figure 1 as the green dots (mean) and bars ($\pm 1$ standard deviation).

The DMC operating on the data transformed by the QFM performs at par with the best classical model operating on the original plain text data. This is true for both the DMC executed on the quantum simulator ("DMC – exact" and "DMC – approx") and the DMC executed on the actual quantum hardware ("DMC – QPU"):

- **DMC – exact:** This is an ideal execution of the DMC algorithm on a quantum simulator using exact density matrices as per QFM expressions (3) and (4).
- **DMC – approx:** This is an approximation based on the expectation values of the first order Pauli operators as per the algorithm outlined in Appendix D executed on a quantum simulator.
- **DMC – QPU:** This is full end-to-end execution of the proposed protocol on the latest generation Rigetti quantum processor (Rigetti Ankaa-3 QPU [26]) as per Section 3 and Algorithm 2.

# 5 Benchmarking Against an Alternative Data Anonymisation Approach

## 5.1 Data Anonymisation with Synthetic Data Generator

In this section we compare performance of the proposed data anonymisation solution based on QFM with an alternative approach based on synthetic data generation. Generative models have been used across a wide range of use cases and their profound impact has been recognised by the Royal Swedish Academy of Sciences that awarded the Nobel Prize in Physics 2024 to John Hopfield and Geoffrey Hinton for foundational discoveries and inventions that enable machine learning with artificial neural networks. The Nobel Committee specifically mentioned the Boltzmann machine – a generative model developed by Hinton using the Hopfield network as the foundation [27].

Note that generative modelling in itself may not be sufficient to guarantee complete data privacy as some off-the-shelf generative models overfit on specific training samples by implicitly memorising them. Therefore, some rudimental information may still be extracted from overfitted models highlighting an important aspect of how anonymisation and overfitting are intertwined.

This motivated us to use a Restricted Boltzmann Machine to provide accurate pattern-based approximations of data distributions. RBMs are generative neural networks that are capable of dealing with the generative model challenges mentioned above [11]. We provide a brief description of RBM architecture, training, and the data generation process in Appendix E. As we shall see below, the RBM is a good choice for producing data samples that conserve the characteristics of the original dataset while preserving privacy. RBMs also have an advantage over other types of generative ML models such as GANs in terms of being able to learn the target distribution from a relatively small number of samples, while GANs require much larger training datasets. Additionally, the autoencoder-based information bottleneck structure of the RBM combined with the fact that it operates on stochastic binary activation units ensures that the generated data points cannot reliably be traced to individual data points in the original data.

## 5.2 Classification Results

The following RBM configuration (scikit-learn Bernoulli RBM [28]) was used for the Wisconsin Breast Cancer dataset [15]:

- Number of binary activation units in the visible layer: $4 \times 9 + 1 = 37$. Each integer feature with values from 1 to 10 can be represented as a 4-digit binary number. The class variable is already binary and can be encoded in a single binary activation unit.
- Number of binary activation units in the hidden layer was set equal to 20 to ensure strong auto-encoder characteristics. The information bottleneck structure coupled with stochastic binary activation units results in strong regularisation properties.
- Mini-batch size: 10
- Learning rate: 0.002
- Number of iterations: 40,000

We can judge how well the RBM can learn a complex multivariate distribution function by comparing the feature importances and a scatter plot (joint distribution) of the two most important features constructed using the original and RBM generated datasets (same number of samples):

- Figure 2 shows that the order of importance is the same for the top three and the bottom two features.
- Figure 3 displays the joint distribution of the top two features, showing that the RBM did a good job learning the dependence structure and the marginal distributions of the features.

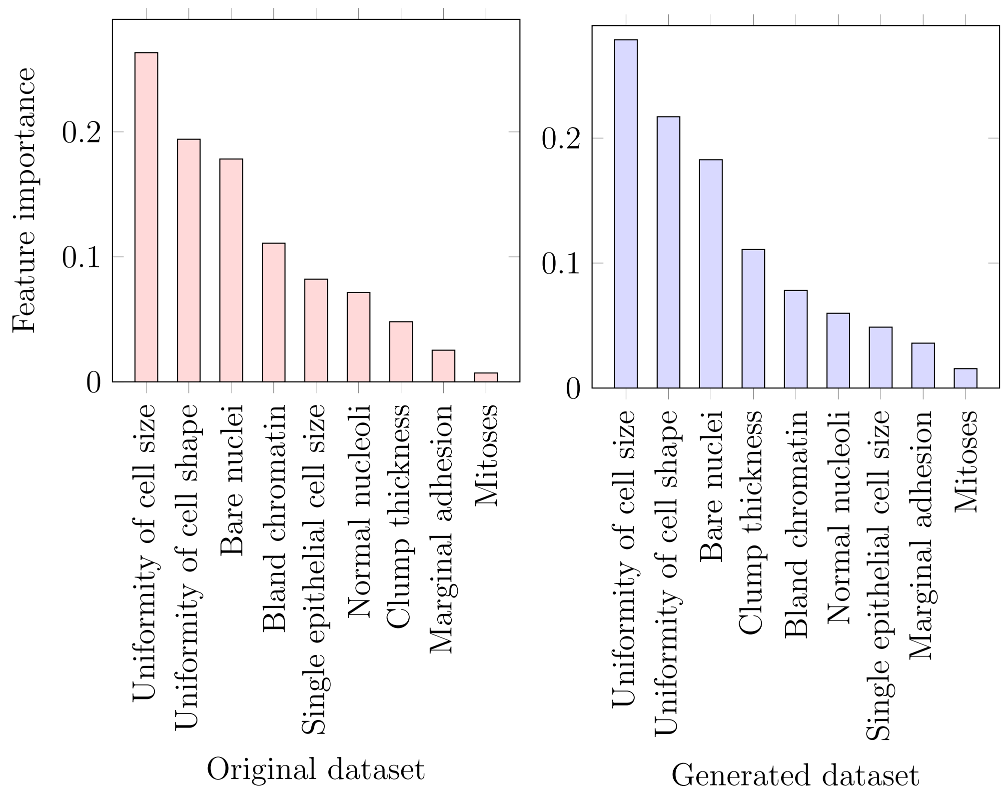


Figure 2: Feature importances for original and generated datasets. The top three and bottom two features are the same for both datasets. This illustrates how the RBM has learned a complex multivariate distribution.

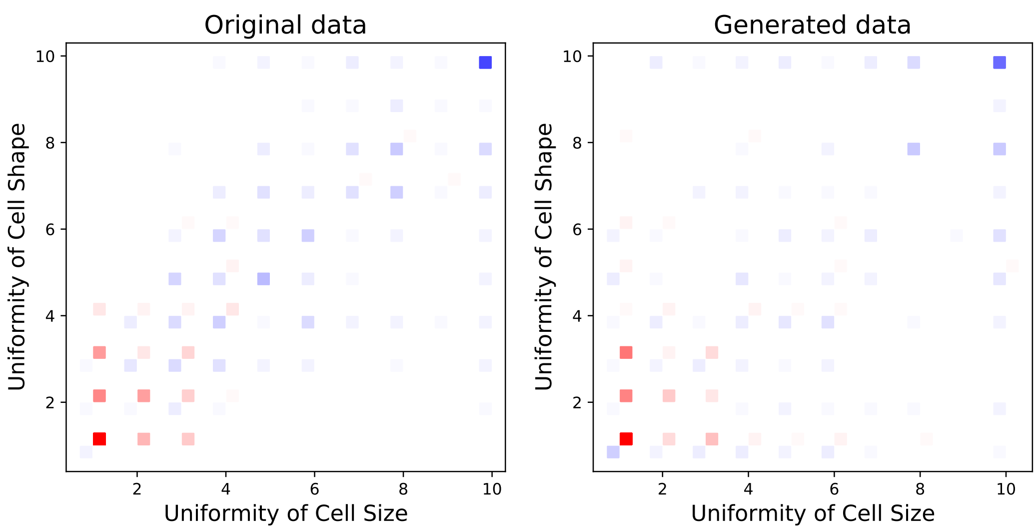


Figure 3: Scatter plots of the two most important features. The red (blue) squares are samples belonging to class 0 (class 1). The colour intensity indicates the frequency of the samples: class 0 samples gravitate towards the lower left corner and class 1 samples are most likely to be found in the upper right corner of the scatter plots.

One of the main advantages of the generative models is their ability to generate an arbitrarily large number of samples drawn from the learned distribution. This may be a crucially important quality when it comes to generating synthetic samples for the subsequent training of various machine learning models. Figure 4 shows out-of-sample F1 scores for the classifiers trained on either synthetic or original datasets. The original in-sample dataset used for training the classifiers consisted of 546 samples. It was also used for training the RBM, with the RBM then generating 1,000 synthetic samples also used to train the same classifiers. As we can clearly see from Figure 4, classifiers trained on synthetic datasets (we used 10 different 80:20 splits into the train/test datasets) display only slightly worse out-of-sample performance in comparison with the same classifiers trained on the original data, well within the margin of error.

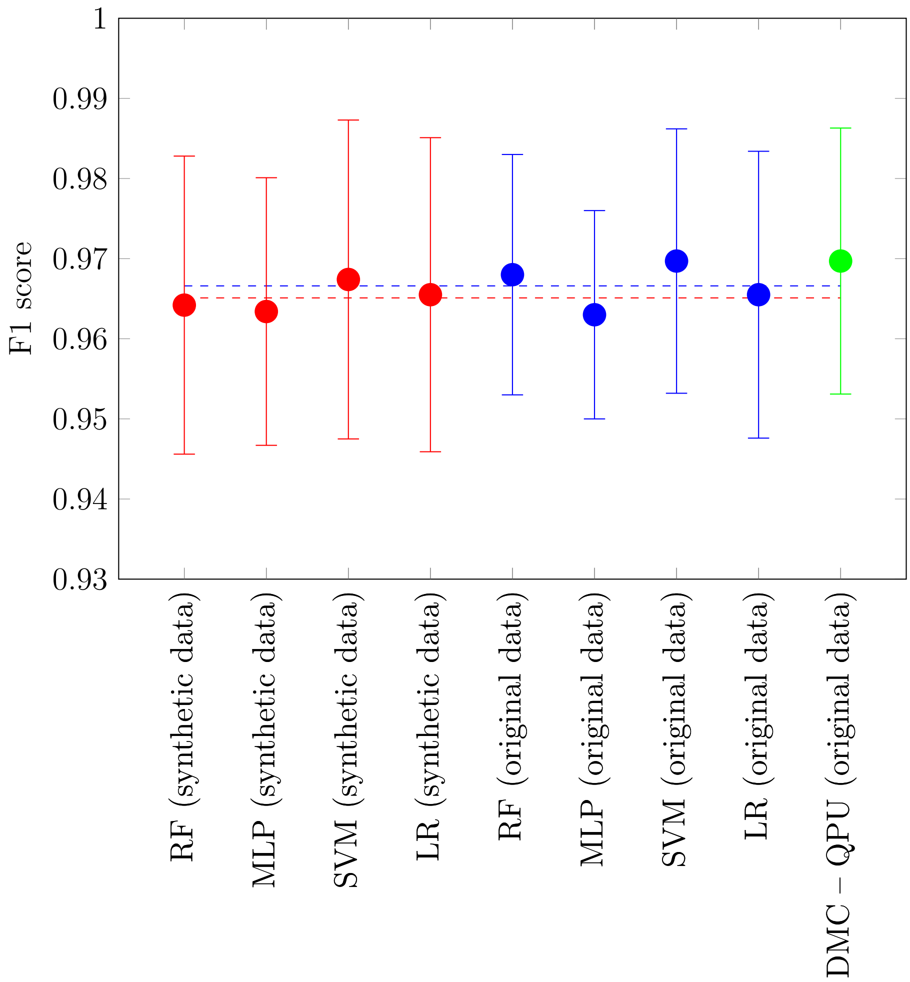


Figure 4: Out-of-sample F1 scores for synthetic (shown in red) and original (shown in blue and green) Wisconsin Breast Cancer datasets. Dots indicate the mean values and error bars indicate ±1 standard deviations. All F1 scores were calculated with "macro" averaging across "benign" and "malignant" classes. The red and blue dashed lines indicate average mean F1 score values across all classical classifiers for the synthetic dataset (red) and original dataset (blue).

This illustrates the expressive power of RBM, that is, its ability to learn complex functions. Using entanglement entropy as a measure of expressive power, it has been proven [29] that the long-range RBM shown in Figure 7 (with every visible layer node linked to every hidden layer node) has larger expressive power than a basic tree-network PQC (see, e.g., Chapter 8 in [14]) but strictly smaller expressive power than the multi-layer PQC of the type shown in Figure 6 when only a polynomial number of parameters is allowed. For systems that exhibit quantum supremacy, a classical model cannot learn to reproduce the statistics unless it uses exponentially scaling resources [30]. This observation is in line with the DMC based on a QFM showing the best result though well within the margin of error.

# 6 Conclusions and Outlook

In this paper we proposed a novel data anonymisation protocol based on the application of a Quantum Feature Map and the Density Matrix Classifier executable on a quantum computer. The main feature of the proposed protocol is the possibility to perform various machine learning tasks directly on the anonymised (encrypted) data.

The expressive power of parameterised quantum circuits used to implement the QFM guarantees preservation of valuable information contained in the original plain text datasets. The DMC is a natural choice of the discriminative model that operates natively on QFMs. The numerical experiments conducted on the popular Wisconsin Breast Cancer dataset demonstrated that the DMC operating on encrypted data performs at par with the standard classical classifiers operating on the original dataset.

We provided further benchmarking against an alternative promising data anonymisation approach based on a synthetic data generator. The classical generative model of choice is the Restricted Boltzmann Machine. The DMC performed well against the RBM, which gives us additional confidence in the capabilities of the proposed data anonymisation protocol.

Interestingly, the proposed protocol appears to be resistant to the noise observed in NISQ-era quantum computing devices. The DMC results obtained on both the QPU and the quantum simulator (an "ideal" quantum computer) are practically indistinguishable. This is due to the fact that the protocol is based on the calculation of expectation values of Pauli operators that averages out random noise, while systematic noise can be mitigated. The improved fidelity of 1-qubit and 2-qubit gates (Rigetti Ankaa-3 QPU) reduces the overall level of noise sufficiently for the proposed protocol to demonstrate its utility on the practical, real-world use case.

We believe that the proposed data anonymisation (encryption) scheme is sufficiently robust and is not vulnerable to either classical or quantum attack. However, a proof of theoretical soundness of the proposed methodology is still outstanding. Follow up research along the lines of [31] should take us closer to understanding the full cryptographic potential of the proposed approach.

Additional research using operator spreading concepts to investigate the distribution of features across qubits, driven by circuit interactions between the qubits, may lead to interesting insights [32, 33, 34]. This could help to quantify further the quality of anonymisation and the capabilities of the machine learning classification. It could assist with the design of quantum circuits which optimise these quantities.

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# A Quantum Feature Map

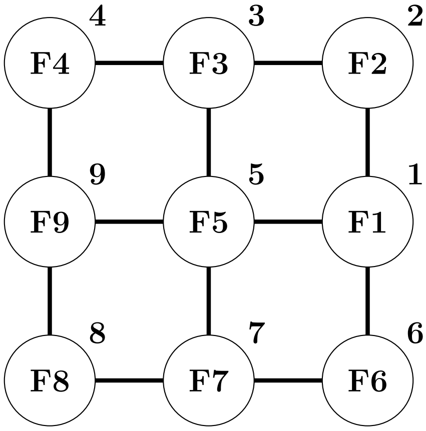


Figure 5: Embedding of a 9-qubit quantum feature map with single qubit feature encoding on a square-grid QPU graph (Ankaa/Novera). Circles indicate computational qubits and lines indicate tunable couplers. Qubits are indexed from 1 to 9 and feature names are shown inside the circles.

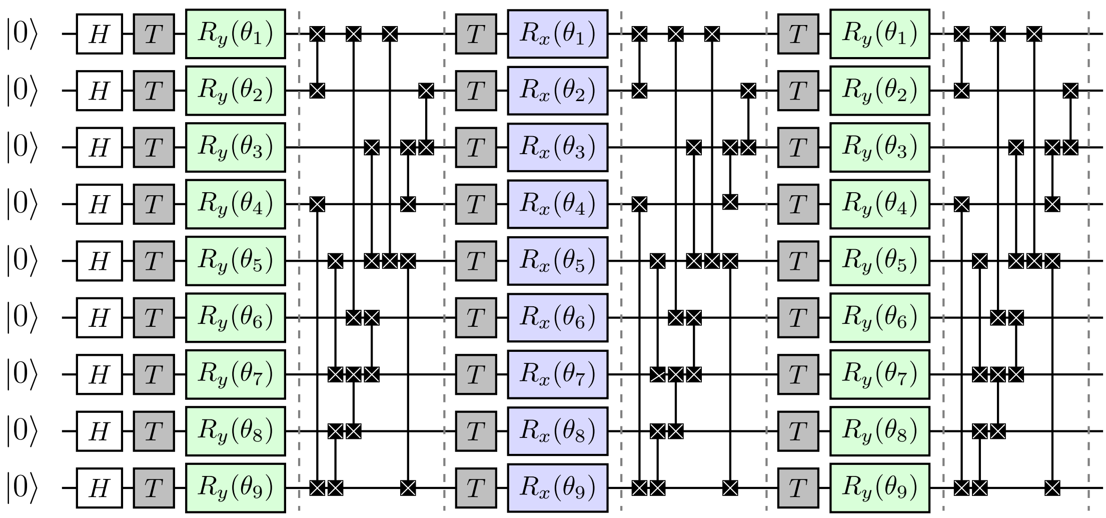

Figure 6: 9-qubit quantum circuit (quantum feature map) with single qubit feature encoding repeated three times. Coloured boxes indicate 1-qubit gates (rotations around the $y$ and $x$ axis) encoding the features and vertical lines indicate 2-qubit gates ($iSWAP$ gates) creating entanglement. $H$ gates (white) are Hadamard gates creating equal superposition of $|0\rangle$ and $|1\rangle$ states. $T$ gates (gray) are non-Clifford phase $\pi/4$ gates adding complexity to the quantum circuit.

## B Feature Encoding

Feature encoding is performed using the *angle encoding* scheme. Each feature $x_i, i = 1, \dots, m$ of a sample $\mathbf{x} := (x_1, \dots, x_m)$ is mapped into the corresponding *rotation angle* – the adjustable parameter of a 1-qubit gate. Each 1-qubit gate operator (mathematically, a $2 \times 2$ unitary matrix) can be seen as a rotation of the qubit state (a point on the unit sphere called the Bloch sphere) around some axis. In practice, and without loss of generality, we work with rotations around the $x$, $y$, and $z$ axis (with $z$-basis being the computational basis). Any point on the Bloch sphere can be reached from any other point on the Bloch sphere by just two rotations around any two orthogonal axes.

Given the dataset $\mathcal{C}$ with the total number of samples equal to the dataset cardinality $[\mathcal{C}]$, for each sample $\mathbf{x}^k, k = 1, \dots, [\mathcal{C}]$, the mapping of the feature $x_i^k$ to the corresponding rotation angle $\theta_i^k$ is performed as follows:

$$\theta_i^k = \alpha\pi \frac{x_i^k - \min\limits_{j=1,\dots,[\mathcal{C}]} x_i^j}{\max\limits_{j=1,\dots,[\mathcal{C}]} x_i^j - \min\limits_{j=1,\dots,[\mathcal{C}]} x_i^j} \,, \tag{5}$$

where $\alpha$ is the range parameter. In our experiments we set $\alpha = 1/4$ and, therefore, we have $\theta_i^k \in [0, \pi/4]$, $\forall\, i, k$. The choice of the range parameter value was dictated by the analysis of the quantum circuit complexity presented in Appendix F.

However, before the feature is mapped into a rotation angle of a 1-qubit gate, it should be pre-processed to even out its distribution. If the feature is distributed more or less evenly, the mapping scheme (5) works well without any pre-processing. But if the feature values are concentrated around the minimum or the maximum values, we need to stretch the distribution up from the minimum or down from the maximum in order to enhance the discriminatory power of a quantum feature map. Intuitively, we can understand it as an attempt to reach as many regions of the Hilbert space as possible and increase the distance between different data points. In our experiments, we adopted the following simple feature pre-processing rule:

- **If** $\text{median} - \text{min} < \frac{1}{4}(\text{max} - \text{min})$ **Then** $x \to x^\gamma$, $\gamma = \frac{1}{2}$.
- **If** $\frac{1}{4}(\text{max} - \text{min}) \le \text{median} - \text{min} \le \frac{3}{4}(\text{max} - \text{min})$ **Then** $x \to x^\gamma$, $\gamma = 1$.
- **If** $\text{median} - \text{min} > \frac{3}{4}(\text{max} - \text{min})$ **Then** $x \to x^\gamma$, $\gamma = \frac{3}{2}$.

## C Classical Benchmarks

The classical benchmarks of choice are four standard classical classifiers which are widely used in data science applications: multi-layer perceptron (MLP), random forest (RF), support vector machine (SVM), and logistic regression (LR). Collectively, they cover all the main classical ML approaches to classification problems. We use the scikit-learn implementation of these classifiers [35] with the set of optimised parameters as shown in Table 4.

| scikit-learn classifier | parameter value |
|---|---|
| Logistic Regression Classifier | solver = 'lbfgs'<br>penalty = 'l2'<br>C = 1<br>max_iter = 5000<br>random_state = 0 |
| MLP Classifier | solver = 'adam'<br>hidden_layer_sizes = (50, 50)<br>activation = 'relu'<br>alpha = 0.1<br>max_iter = 5000<br>random_state = 0 |
| Random Forest Classifier | criterion = 'entropy'<br>n_estimators = 500<br>random_state = 0 |
| Support Vector Classifier | kernel = 'rbf'<br>C = 1<br>random_state = 0 |

Table 4: scikit-learn classifiers – optimised model parameters. All other model parameters were fixed at their default values [35].

Similar to the feature standardisation for DMC, where all features are mapped to the same rotation angle interval $[0, \alpha\pi]$ as per expression (5), classical classifiers such as MLP, SVM and LR also work with standardised features. For our experiments we use the scikit-learn StandardScaler class [36], which standardises features by removing their mean values and scaling them to unit variance.

# D Estimation of Frobenius Distance on a Quantum Computer

Consider the space of linear operators $\mathcal{H}_{\mathcal{O}}$ with dimension $D_{\mathcal{O}} = D^2 = (2^n)^2$ acting in the Hilbert space $\mathcal{H}$ of a quantum system with dimension $D = 2^n$. A distance squared $d^2$ between two Hermitian operators $p$ and $q$ can be derived from a linear product $(\cdot, \cdot)$ defined in $\mathcal{H}_{\mathcal{O}}$:

$$d^2(p, q) = \|p - q\|^2 = (p - q, p - q).$$

If $\{B_i\}$ is a complete orthonormal set in $\mathcal{H}_{\mathcal{O}}$, then we can express $p$ and $q$ as vectors in $\mathbb{C}^{D_{\mathcal{O}}}$ with components $p_i := (B_i, p)$ and $q_i := (B_i, q)$ respectively. The Pauli operators

$$\{V_i := V_i^1 \otimes \ldots \otimes V_i^n\}, \quad V_i^j = \{I, X, Y, Z\}, \tag{6}$$

corresponding to the $j$-th single-qubit identity and Pauli operators, are an orthogonal set with the choice of linear product

$$(L, M) = \mathrm{Tr}[L^\dagger M],$$

which induces the Frobenius norm. The distance squared $d^2$ can be expressed as the Euclidean distance in $\mathbb{C}^{D_\mathcal{O}}$:

$$d^2(p,q) = \sum_{i=1}^{D_\mathcal{O}} |p_i - q_i|^2 = \|\mathbf{p} - \mathbf{q}\|_E^2,$$

with column vectors

$$\mathbf{p} := (p_1, \dots, p_{D_\mathcal{O}})^\top \quad \text{and} \quad \mathbf{q} := (q_1, \dots, q_{D_\mathcal{O}})^\top.$$

Fixing the normalisation of the operators $\{V_i\}$, i.e., choosing $B_i = V_i/\sqrt{D}$ and $p$ and $q$ representing density matrices, the elements $p_i$ and $q_i$ become real numbers that can be expressed as

$$p_i = \frac{1}{\sqrt{D}} \mathrm{Tr}[V_i p] = \frac{1}{\sqrt{D}} \left\langle V_i \right\rangle_p$$

and

$$q_i = \frac{1}{\sqrt{D}} \mathrm{Tr}[V_i q] = \frac{1}{\sqrt{D}} \left\langle V_i \right\rangle_q,$$

and therefore

$$d^2(p,q) = \frac{1}{D} \sum_{i=1}^{D_\mathcal{O}} |\mathrm{Tr}[V_i p] - \mathrm{Tr}[V_i q]|^2 = \frac{1}{D} \sum_{i=1}^{D_\mathcal{O}} |\left\langle V_i \right\rangle_p - \left\langle V_i \right\rangle_q|^2.$$

The Frobenius distance as defined in (4) can be written as a Euclidean distance between vectors of expectation values of Paulis

$$d_F^2(p,q) = \frac{1}{4D} \sum_{i=1}^{D_\mathcal{O}} |\left\langle V_i \right\rangle_p - \left\langle V_i \right\rangle_q|^2 = \frac{1}{4D} \|\mathbf{P} - \mathbf{Q}\|_E^2,$$

where

$$\mathbf{P} := (\left\langle V_1 \right\rangle_p, \dots, \left\langle V_{D_\mathcal{O}} \right\rangle_p)^\top \quad \text{and} \quad \mathbf{Q} := (\left\langle V_1 \right\rangle_q, \dots, \left\langle V_{D_\mathcal{O}} \right\rangle_q)^\top.$$

The complete basis set of operators $V_i$ consists of $4^n$ elements. This makes working with the complete basis set impractical or even impossible for large $n$. Therefore, we choose a subset of $L$ operators $V_i$ from the full set and approximate the sum on the expectation values of all Pauli operators with the sum on the subset of $L$ Pauli operators:

$$\frac{1}{4D} \sum_{i=1}^{L} |\left\langle V_i \right\rangle_p - \left\langle V_i \right\rangle_q|^2 \leq \frac{1}{4D} \sum_{i=1}^{D_\mathcal{O}} |\left\langle V_i \right\rangle_p - \left\langle V_i \right\rangle_q|^2.$$

For example, $L$ can be the set of all first order Pauli operators. For many practical applications we can ignore the higher order tensor products of Pauli operators since their expectation values tend to make negligibly small contribution.

Thus, we can define the Frobenius distance estimator as

$$\tilde{d}_F(p,q) := \frac{1}{2}\sqrt{\frac{\sum_{i=1}^{L} |\langle V_i \rangle_p - \langle V_i \rangle_q|^2}{D}}.$$

If we consider the subspace of $\mathcal{H}_{\mathcal{O}}$ spanned by the $L$ Pauli operators, then

$$\tilde{d}_F(p,q) := \frac{1}{2}\sqrt{\frac{\|\mathbf{P}_L - \mathbf{Q}_L\|_E^2}{D}},$$

where

$$\mathbf{P}_L := (\langle V_1 \rangle_p, \ldots, \langle V_L \rangle_p)^\top \quad \text{and} \quad \mathbf{Q}_L := (\langle V_1 \rangle_q, \ldots, \langle V_L \rangle_q)^\top.$$

We can interpret this as the exact Frobenius distance with a quantum feature map that does not map to the full density matrix, but to vectors of components of the full density matrix on a Pauli basis.

Algorithm 2 provides detailed instructions for obtaining the Frobenius distance estimate on a quantum computer between the density matrices representing two datasets where we restrict ourselves to using the first order Pauli operators only.

**Algorithm 2** Frobenius distance estimator for two datasets

**Result:** Frobenius distance.

1: Design $n$-qubit quantum circuit (quantum feature map) with layers of one-qubit gates (encoding sample) alternating with layers of two-qubit gates (creating entanglement). By default, the range of rotation angles should be set at $[0, \pi/m]$, where $m$ is the number of data encoding layers.

2: Prepare two datasets ($A$ and $B$) – calculate features, standardise features, etc.

3: For dataset $A$:

- For each sample $i = 1, \ldots, [A]$, run the quantum circuit $N$ times measuring Pauli $Z_k$ on all quantum registers $k = 1, \ldots, n$ ($+1$ if we measure $|0\rangle$ and $-1$ if we measure $|1\rangle$). Calculate expectation value $\langle Z_k \rangle_i^A$ as the average value of measured $Z_k$ across all quantum circuit runs.
- For each sample $i = 1, \ldots, [A]$, place $H$ gate on **all** quantum registers before measurement and run the quantum circuit $N$ times measuring Pauli $X_k$ on all quantum registers $k = 1, \ldots, n$ ($+1$ if we measure $|0\rangle$ and $-1$ if we measure $|1\rangle$). Calculate expectation value $\langle X_k \rangle_i^A$ as the average value of measured $X_k$ across all quantum circuit runs.
- For each sample $i = 1, \ldots, [A]$, place $HS^\dagger$ gates on **all** quantum registers before measurement and run the quantum circuit $N$ times measuring Pauli $Y_k$ on all quantum registers $k = 1, \ldots, n$ ($+1$ if we measure $|0\rangle$ and $-1$ if we measure $|1\rangle$). Calculate expectation value $\langle Y_k \rangle_i^A$ as the average value of measured $Y_k$ across all quantum circuit runs.

4: For dataset $B$:

- For each sample $j = 1, \ldots, [B]$, run the quantum circuit $N$ times measuring Pauli $Z_k$ on all quantum registers $k = 1, \ldots, n$ ($+1$ if we measure $|0\rangle$ and $-1$ if we measure $|1\rangle$). Calculate expectation value $\langle Z_k \rangle_j^B$ as the average value of measured $Z_k$ across all quantum circuit runs.
- For each sample $j = 1, \ldots, [B]$, place $H$ gate on **all** quantum registers before measurement and run the quantum circuit $N$ times measuring Pauli $X_k$ on all quantum registers $k = 1, \ldots, n$ ($+1$ if we measure $|0\rangle$ and $-1$ if we measure $|1\rangle$). Calculate expectation value $\langle X_k \rangle_j^B$ as the average value of measured $X_k$ across all quantum circuit runs.
- For each sample $j = 1, \ldots, [B]$, place $HS^\dagger$ gates on **all** quantum registers before measurement and run the quantum circuit $N$ times measuring Pauli $Y_k$ on all quantum registers $k = 1, \ldots, n$ ($+1$ if we measure $|0\rangle$ and $-1$ if we measure $|1\rangle$). Calculate expectation value $\langle Y_k \rangle_j^B$ as the average value of measured $Y_k$ across all quantum circuit runs.

5: We now have two arrays: $[A] \times 3n$ array of expectation values of first order Paulis for dataset $A$ and $[B] \times 3n$ array of expectation values of first order Paulis for dataset $B$. For every column of these arrays we calculate the average expectation values across all samples in the datasets:

$$\overline{\langle X_1 \rangle^A} = \frac{1}{[A]} \sum_{i=1}^{[A]} \langle X_1 \rangle_i^A, \ldots; \quad \overline{\langle X_1 \rangle^B} = \frac{1}{[B]} \sum_{j=1}^{[B]} \langle X_1 \rangle_j^B, \ldots$$

6: We now have two $3n$-dimensional column vectors for datasets $A$ and $B$:

$$\left(\overline{\langle X_1 \rangle^A}, \overline{\langle X_2 \rangle^A}, \ldots\right)^\top; \quad \left(\overline{\langle X_1 \rangle^B}, \overline{\langle X_2 \rangle^B}, \ldots\right)^\top.$$

The next step is to calculate the column vector of differences:

$$\mathbf{c} = \left(\overline{\langle X_1 \rangle^A} - \overline{\langle X_1 \rangle^B}, \overline{\langle X_2 \rangle^A} - \overline{\langle X_2 \rangle^B}, \ldots\right)^\top.$$

7: Calculate the estimate of the Frobenius distance:

$$\tilde{d}_F(A, B) = \frac{1}{2} \sqrt{\frac{\mathbf{c}^\top \mathbf{c}}{2^n}}.$$

# E Restricted Boltzmann Machine

The Restricted Boltzmann Machine is a shallow two-layer neural network that operates on stochastic binary activation units. The network forms a bipartite graph connecting stochastic binary inputs (visible units) to stochastic binary feature detectors (hidden units) with no connections between the units within the same layer, as shown in Figure 7 [37].

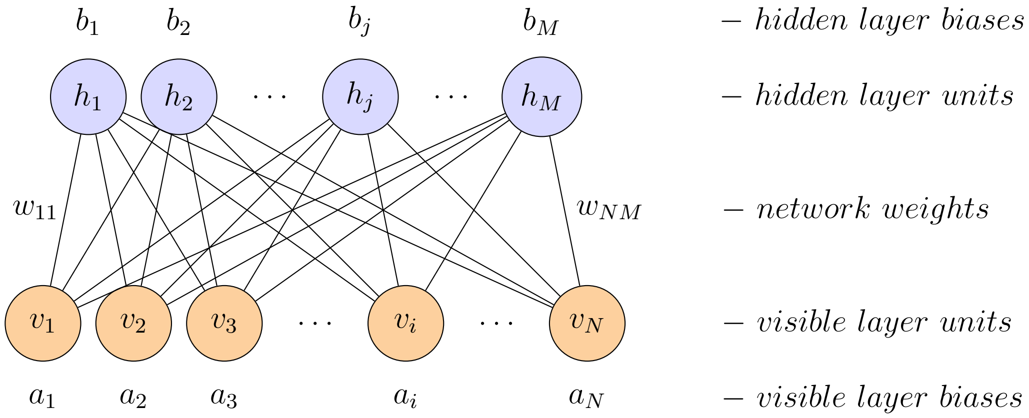


Figure 7: Schematic representation of an RBM.

Only the visible layer of the network is exposed to the training dataset and its inputs $\mathbf{v} := (v_1, \ldots, v_N)$ flow through the network (forward pass) to the hidden layer, where they are aggregated and added to the hidden layer biases $\mathbf{b} := (b_1, \ldots, b_M)$. The hidden layer sigmoid activation function $\sigma(x) := 1/(1 + \exp(-x))$ converts aggregated inputs into probabilities. Each hidden unit then "fires" randomly and outputs a $\{0, 1\}$ Bernoulli random variable with the associated probabilities:

$$\mathbb{P}(h_j = 1|\mathbf{v}) = \sigma\left(b_j + \sum_{i=1}^{N} w_{ij}v_i\right) \quad \text{and} \quad \mathbb{P}(h_j = 0|\mathbf{v}) = 1 - \sigma\left(b_j + \sum_{i=1}^{N} w_{ij}v_i\right).$$

The outputs from the hidden layer $\mathbf{h} := (h_1, \ldots, h_M)$ then flow back (backward pass) to the visible layer, where they are aggregated and added to the visible layer biases $\mathbf{a} := (a_1, \ldots, a_N)$. Similar to the hidden layer, the visible layer sigmoid activation function first translates aggregated inputs into probabilities and then into Bernoulli random variables:

$$\mathbb{P}(v_i = 1|\mathbf{h}) = \sigma\left(a_i + \sum_{j=1}^{M} w_{ij}h_j\right) \quad \text{and} \quad \mathbb{P}(v_i = 0|\mathbf{h}) = 1 - \sigma\left(a_i + \sum_{j=1}^{M} w_{ij}h_j\right).$$

Therefore, every unit communicates at most one bit of information. This is especially important for the hidden units since this feature implements the information bottleneck structure, which acts as a strong regulariser [38]. The hidden layer of the network can learn the low-dimensional probabilistic representation of the dataset if the network is organised and trained as an autoencoder.

Figure 8 illustrates the binary representation of an input signal that enters the network through the visible layer. The number of activation units in the visible layer is determined by the number of features we have to encode and the desired precision of their binary representation. For example, if our sample consists of two continuous features encoded, respectively, as an $n$-digit and an $m$-digit binary numbers, then the total number of activation units in the visible layer is $n+m$.

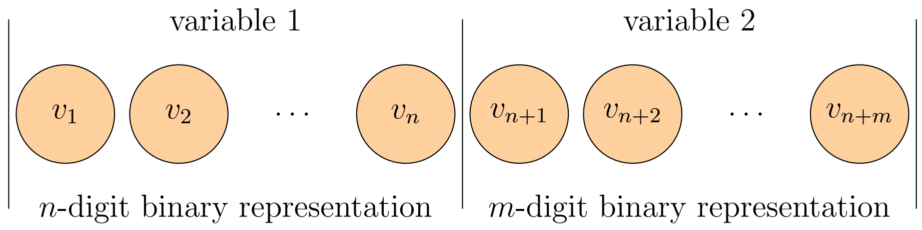


Figure 8: Schematic binary encoding of continuous variables.

The network learns the probability distribution $\mathbb{P}(\mathbf{v}, \mathbf{h})$ of the configurations of visible and hidden activation units – the Boltzmann distribution – by trying to reconstruct the inputs from the training dataset (visible unit values) through finding an optimal set of the network weights and biases:

$$\mathbb{P}(\mathbf{v}, \mathbf{h}) = \frac{1}{Z} \mathrm{e}^{-E(\mathbf{v}, \mathbf{h})},$$

where the energy function reads

$$E(\mathbf{v}, \mathbf{h}) = -\sum_{i=1}^{N} a_i v_i - \sum_{j=1}^{M} b_j h_j - \sum_{i=1}^{N} \sum_{j=1}^{M} w_{ij} v_i h_j.$$

Here, $Z$ is the partition function:

$$Z = \sum_{\mathbf{v}, \mathbf{h}} \mathrm{e}^{-E(\mathbf{v}, \mathbf{h})}.$$

We are usually interested in learning either the probability distribution of the visible layer configurations if we want to generate new samples that would have the same statistical properties as the original training dataset, or in learning the probability distribution of the hidden layer configurations if we want to build a deep neural network where the RBM layer performs the feature extraction and dimensionality reduction function. The probabilities of the visible (hidden) states are given by summing over all possible hidden (visible) vectors:

$$\mathbb{P}(\mathbf{v}) = \frac{1}{Z} \sum_{\mathbf{h}} \mathrm{e}^{-E(\mathbf{v}, \mathbf{h})} \quad \text{and} \quad \mathbb{P}(\mathbf{h}) = \frac{1}{Z} \sum_{\mathbf{v}} \mathrm{e}^{-E(\mathbf{v}, \mathbf{h})}.$$

The most popular training algorithm for RBM, the *k-step Contrastive Divergence*, was proposed by Hinton [38]. The algorithm aims to maximise the log-likelihood of a training vector, i.e., to find such network weights and biases that the energy function $E$ is minimised for the samples from the training dataset (smaller value of energy corresponds to larger probability of a configuration). The interested reader can also find an excellent introduction to the training of RBMs in the work by Fischer and Igel [39].

Once fully trained, the network can be used to generate new samples from the learned distribution. For example, the RBM can be used as a market generator that produces new market scenarios in the form of new synthetic samples drawn from the multivariate distribution of market risk factors encoded in the network weights and biases.

The first step is the generation of a random input where each visible unit is initialised with a randomly generated binary variable. The second step is performing a large number of forward and backward passes between the visible and the hidden layers, until the system reaches a state of *thermal equilibrium*, a state where the initial random vector is transformed into a sample from the learned distribution. The number of cycles needed to reach the state of thermal equilibrium is problem dependent and is a function of network architecture and network parameters (weights and biases). In some cases, the generation of independent samples requires between $10^3$ and $10^4$ forward and backward passes through the network [40]. The final step is the readout from the visible layer, which gives us a bitstring, encoding a sample from the target distribution.

# F Magic in Parameterised Quantum Circuits

The question of what gives a PQC its expressive power is a deeply non-trivial one. The expressive power cannot be explained by the quantum mechanical properties of superposition and entanglement alone. It is possible to create examples where a classical model implements a feature map into a Hilbert space of infinite dimensions (e.g., Gaussian kernel), and where a highly entangled quantum state (e.g., the GHZ state) is lacking complexity and can be efficiently simulated classically.

Therefore, to address the question of the expressive power of a QML model we need to consider the complexity of the quantum states that can be obtained using that model. This approach can be formalised with the help of a quantity called *magic* that measures the hardness of classical simulation of the given quantum states (either pure or mixed). Quantum feature maps with higher magic are those that are hard to simulate classically. Thus, there may be a clear gap in the expressive power of the quantum and classical models, unless classical models are given an exponentially larger amount of resources.

Following the formalism of the quantum feature map introduced in Section 2, a sample $\mathbf{x}^k$, $k = 1, \ldots, N$, is mapped into an operator $\varphi(\mathbf{x}^k)$ via a unitary transformation $U(\mathbf{x}^k)$ applied to the all-zero initial state $|0\rangle^{\otimes n}$ as per expression (2). Without loss of generality, let us assume that we are working with a binary classification problem with classes $\mathcal{C}_1$ and $\mathcal{C}_2$ with $[\mathcal{C}_1]$ and $[\mathcal{C}_2]$ elements respectively ($[\mathcal{C}_1] + [\mathcal{C}_2] = N$) and define the following density matrices encoding the datasets:

$$\rho = \frac{1}{N} \sum_{k=1}^{N} \varphi(\mathbf{x}^k), \quad \rho_1 = \frac{1}{[\mathcal{C}_1]} \sum_{k \in \mathcal{C}_1} \varphi(\mathbf{x}^k), \quad \rho_2 = \frac{1}{[\mathcal{C}_2]} \sum_{k \in \mathcal{C}_2} \varphi(\mathbf{x}^k). \tag{7}$$

Further, we can generalise the quantum feature map $\varphi(\mathbf{x}^k)$ by replacing $|0\rangle\langle 0|$ in expression (2) with generic stabiliser initial state $\sigma$:

$$\varphi(\mathbf{x}^k) = U(\mathbf{x}^k)\sigma U^\dagger(\mathbf{x}^k). \tag{8}$$

Then, using the density matrices and the QFM given by expressions (7) and (8), we can define the following quantities [41]:

- *Generalisation capacity* is defined as

$$\mathcal{G} = \sqrt{\mathrm{Tr}[\rho^2]} \equiv \|\rho^2\|_{S_2}, \tag{9}$$

  with $S_2$ the Schatten 2-norm.

- *Approximation capacity* is defined as

$$\mathcal{A} = \|p_1\rho_1 - p_2\rho_2\|_{S_1}, \tag{10}$$

  with $S_1$ the Schatten 1-norm, $p_1 = [\mathcal{C}_1]/N$, $p_2 = [\mathcal{C}_2]/N$. If we are interested in the feature map transformation of the data without assuming any particular learning model (i.e., how the class labels are assigned by the model down the machine learning pipeline), then we can define approximation as

$$\mathcal{A} = \frac{1}{\mathcal{P}} \sum_{\mathcal{P}} \|p_{1_\mathcal{P}}\rho_{1_\mathcal{P}} - p_{2_\mathcal{P}}\rho_{2_\mathcal{P}}\|_{S_1} \equiv \mathbb{E}_\mathcal{P}\left[\|p_{1_\mathcal{P}}\rho_{1_\mathcal{P}} - p_{2_\mathcal{P}}\rho_{2_\mathcal{P}}\|_{S_1}\right],$$

  with $\mathcal{P}$ representing all possible permutations of the labels in the classes.

- *Magic* is defined as

$$\mathcal{M} = \frac{1}{N}\sum_{k=1}^{N} \mathcal{D}(\varphi(\mathbf{x}^k)),$$

  with $\mathcal{D}$ stabiliser norm defined as

$$\mathcal{D}(\varphi(\mathbf{x}^k)) = \frac{1}{D}\sum_{i=1}^{D^2} |\mathrm{Tr}[V_i\varphi(\mathbf{x}^k)],$$

  where $V_i$ are the Pauli operators as defined in expression (6), $D = 2^n$ with $n$ number of qubits in the quantum feature map, and the sum goes over all $4^n$ possible Pauli operators.

It was proposed in [41] that the achievable tradeoff between generalisation capacity and approximation capacity is controlled by the magic of the given quantum feature map:

$$\mathcal{G} \times \mathcal{A} \leq \frac{c}{\sqrt{N}}\mathcal{M},$$

with $c \approx 2/\sqrt{\pi}$ for $N \gg 1$. Therefore, our task is twofold. First, we should construct a high magic PQC ansatz for the quantum feature map – a quantum circuit that is structurally complex. Second, we should optimise any available hyperparameters in a way that maximises the magic. We illustrate these points on the Wisconsin Breast Cancer dataset.

The PQC we used for the data anonymisation and which is shown in Figure 6 can be seen as a high magic ansatz since it contains non-Clifford $T$ gates on all quantum registers in every layer of 1-qubit gates. Since quantum circuits consisting of only Clifford gates can be efficiently simulated classically [42], adding non-Clifford gates increases complexity and, therefore, the magic of the corresponding QFM.

For comparison purposes, we can construct a less complex version of essentially the same ansatz as shown in Figure 9. We reduce complexity by removing all $T$ gates and replacing 2-qubit $iSWAP$ gates with $CZ$ gates. We note that both $iSWAP$ and $CZ$ are Clifford gates, but it seems that in the context of our ansatz the $iSWAP$ gate (a 2-qubit $XX + YY$ interaction) provides a higher degree of complexity.

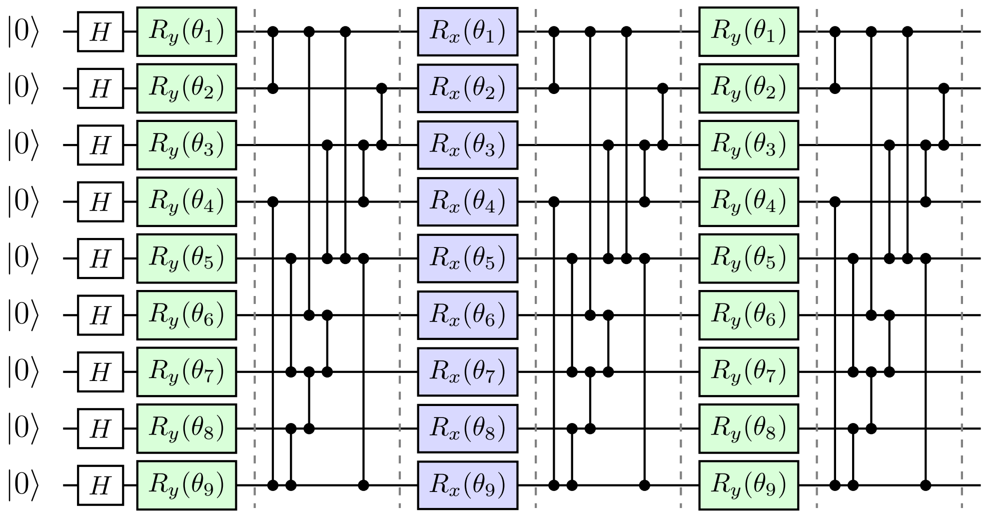


Figure 9: Lower magic 9-qubit quantum circuit ansatz (quantum feature map) with single qubit feature encoding repeated three times. Coloured boxes indicate 1-qubit gates (rotations around the $y$ and $x$ axis) encoding the features and vertical lines indicate 2-qubit gates ($CZ$ gates) creating entanglement. $H$ gates (white) are Hadamard gates creating equal superposition of $|0\rangle$ and $|1\rangle$ states.

The only hyperparameter that can have an impact on the circuit magic is the range parameter $\alpha$ in the angle encoding scheme we use to encode the samples. Therefore, we can investigate how the approximation-generalisation tradeoff depends on both the ansatz and the range parameter by doing a systematic evaluation of both quantities for all reasonable values of the range parameter for the high magic and low magic ansätze.

In Figures 10-16 we show the results of this analysis which were obtained for ten random splits of the dataset into train and test sets in proportion 80:20 – exactly the same train/test split as we applied in Section 4.3. The $\mathcal{G}$ and $\mathcal{A}$ were estimated as per expressions (9) and (10). The impact on DMC performance was captured by F1 score – in line with the results reported in Figure 1 for the high magic ansatz and a single value of the range parameter ($\alpha = 0.25$).

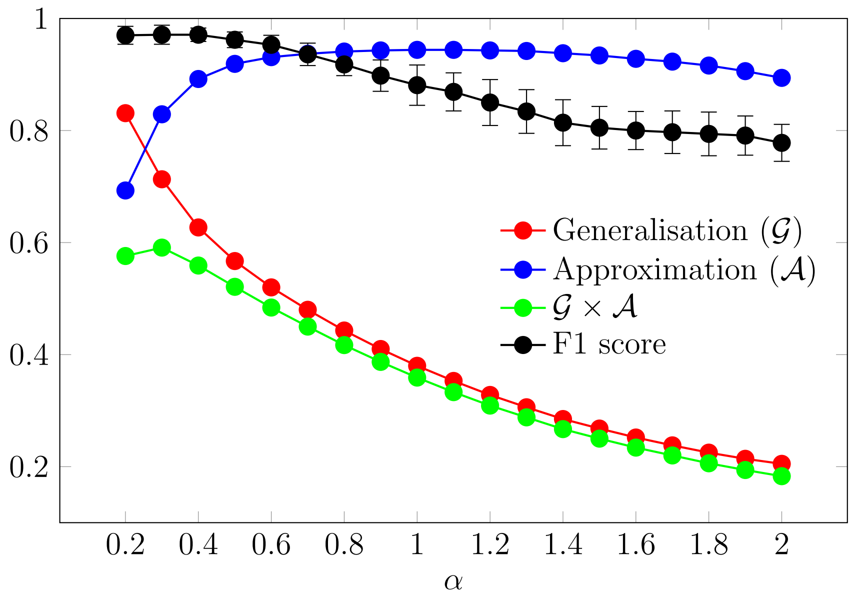


Figure 10: Results when using the higher magic quantum circuit ansatz with 2-qubit $iSWAP$ gates and with additional 1-qubit $T$ gates for the Wisconsin Breast Cancer dataset. The circuit magic and, therefore, the overall model performance depends on the range parameter $\alpha$ in the chosen data encoding scheme. Dots indicate mean values and error bars indicate $\pm 1$ standard deviation.

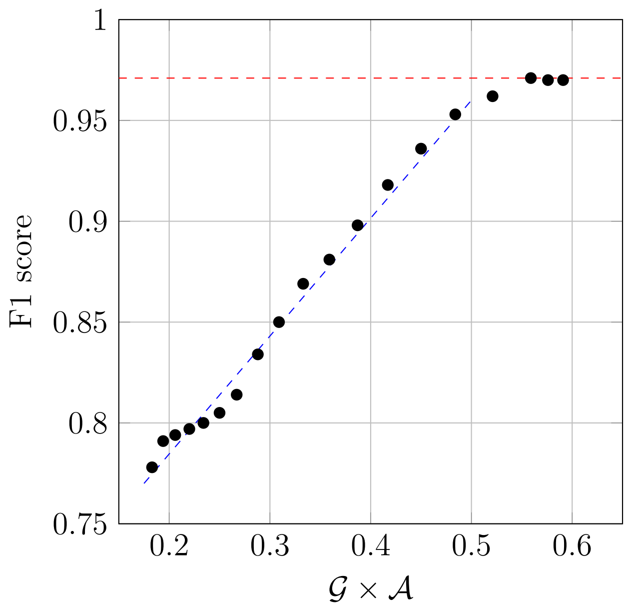


Figure 11: Out-of-sample F1 score as a function of $\mathcal{G} \times \mathcal{A}$ for the higher magic quantum circuit ansatz (Wisconsin Breast Cancer dataset). The chart shows near linear dependence of the model performance as measured by F1 score on $\mathcal{G} \times \mathcal{A}$ until the F1 score reaches the maximum determined by the amount of information that can be extracted from the training dataset.

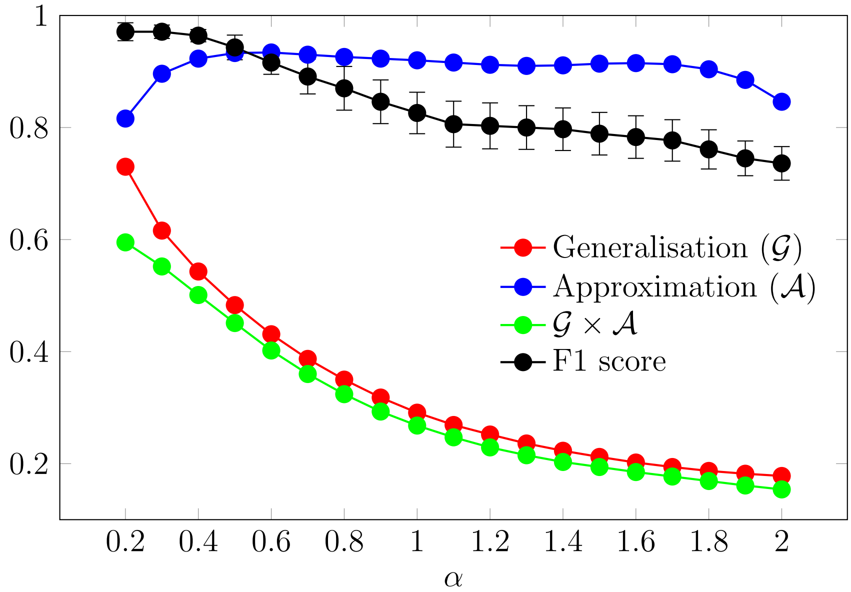


Figure 12: Results when using the lower magic quantum circuit ansatz with 2-qubit $CZ$ gates and without 1-qubit $T$ gates for the Wisconsin Breast Cancer dataset. The circuit magic and, therefore, the overall model performance depends on the range parameter $\alpha$ in the chosen data encoding scheme. Dots indicate mean values and error bars indicate $\pm 1$ standard deviation.

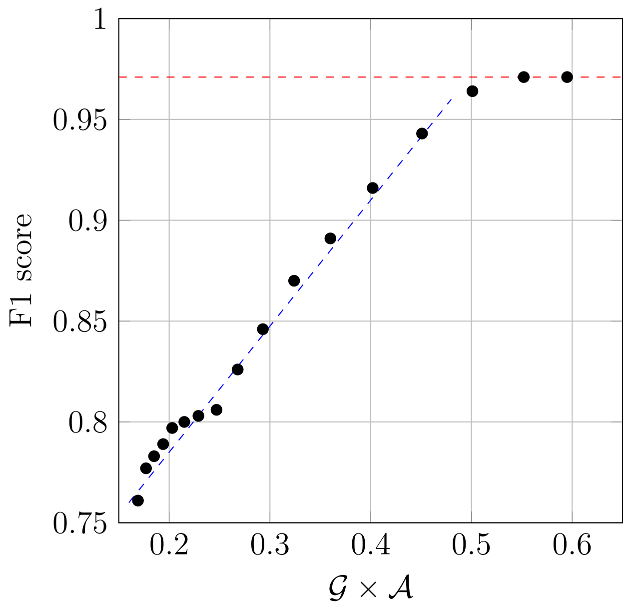


Figure 13: Out-of-sample F1 score as a function of $\mathcal{G} \times \mathcal{A}$ for the lower magic quantum circuit ansatz (Wisconsin Breast Cancer dataset). The chart for the lower magic ansatz also shows near linear dependence of the model performance as measured by F1 score on $\mathcal{G} \times \mathcal{A}$ until the F1 score reaches the maximum determined by the amount of information that can be extracted from the training dataset. However, the density of points in the region of high $\mathcal{G} \times \mathcal{A}$ values is lower in comparison with the same plot for the higher magic ansatz.

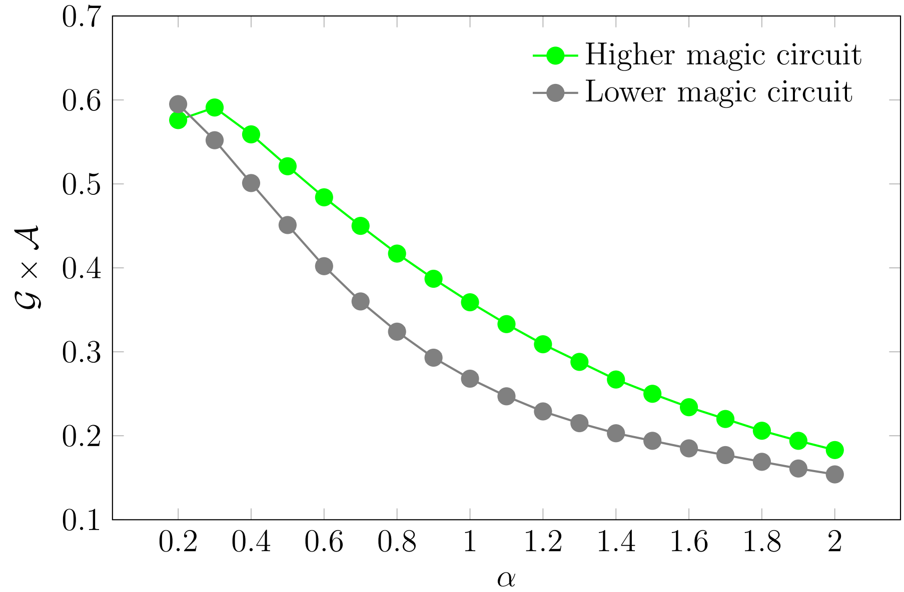


Figure 14: $\mathcal{G} \times \mathcal{A}$ of the higher and lower magic quantum circuit ansätze for the Wisconsin Breast Cancer dataset as a function of the range parameter $\alpha$ in the chosen data encoding scheme.

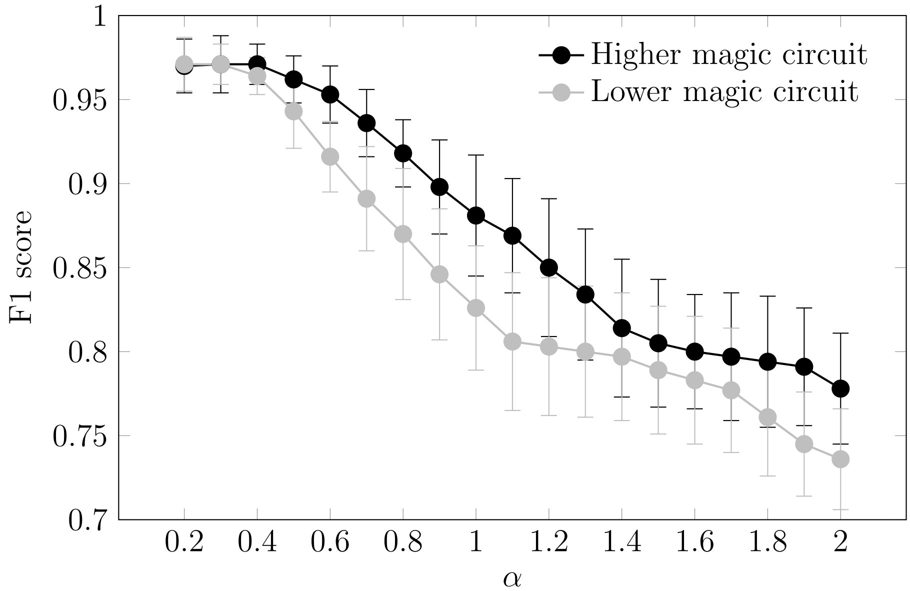


Figure 15: Out-of-sample F1 score of the higher and lower magic quantum circuit ansätze for the Wisconsin Breast Cancer dataset as a function of the range parameter $\alpha$ in the chosen data encoding scheme. Dots indicate mean values and error bars indicate $\pm 1$ standard deviation.

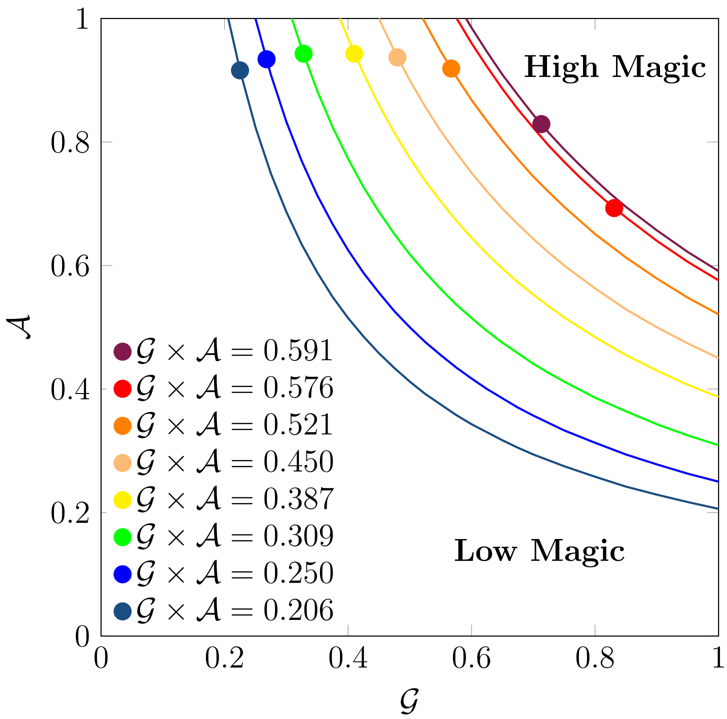


Figure 16: Improvement in approximation-generalisation tradeoff through magic for the Wisconsin Breast Cancer dataset. The dots indicate specific combinations of approximation and generalisation values achieved with the help of parameterised quantum circuit shown in Figure 6. The lines are isolines of equal values of the product of approximation and generalisation. The dots are travelling through the phase space $(\mathcal{G}, \mathcal{A})$ in response to the change in the model hyperparameter. The F1 scores corresponding to the red and dark red dots are practically identical due to the closeness of their isolines, despite the fact that these dots are relatively far from each other in the phase space.

The results of our analysis suggest that the optimal value of the range parameter $\alpha$ is somewhere between 0.2 and 0.3. Therefore, for the purposes of our numerical experiments, we have chosen $\alpha = 0.25$.

Finally, Figure 17 shows the stabiliser norm magic as a function of the data encoding range parameter for the high magic (Figure 6) and low magic (Figure 9) circuits. Not surprisingly, the high magic curve goes much higher than the low magic one. What deserves special attention is magic's cyclical dependency on the range parameter $\alpha$. The magic dips for the range parameter values 0.5, 1, 1.5, and 2. These values correspond to the rotation angle ranges of $[0, \frac{1}{2}\pi]$, $[0, \pi]$, $[0, \frac{3}{2}\pi]$, and $[0, 2\pi]$. The local peaks are observed for the range parameter values 0.25, 0.75, 1.25, and 1.75. These values correspond to the rotation angle ranges of $[0, \frac{1}{4}\pi]$, $[0, \frac{3}{4}\pi]$, $[0, 1\frac{1}{4}\pi]$, and $[0, 1\frac{3}{4}\pi]$. The intuitive explanation is that the quantum circuit magic increases with the addition of non-Clifford $T$ gates (phase shift by angle $\frac{1}{4}\pi$) and drops when rotation angle becomes $\frac{1}{2}\pi$ (Clifford $S$ gate). This further confirms that $\alpha = 0.25$ is the right choice of the range parameter value.

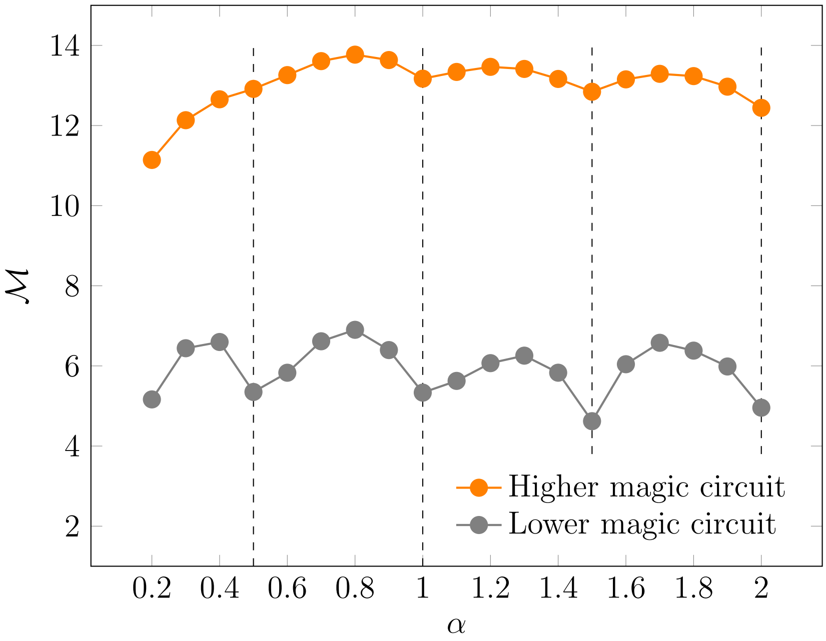


Figure 17: Stabiliser norm magic of the higher and lower magic quantum circuit ansätze for the Wisconsin Breast Cancer dataset as a function of the range parameter $\alpha$ in the chosen data encoding scheme.